

Foundation (Jalapeno)

Q1. Which system of linear inequalities has the solution region shown?

- (A) $x + y \geq 2$ and $x - y \leq 1$
 (B) $x + y \leq 2$ and $x - y \geq 1$
 (C) $x + y \geq 2$ and $x - y \geq 1$
 (D) $x + y \leq 2$ and $x - y \leq 1$
 (E) $x + y \geq 1$ and $x - y \leq 2$

Q2. In a space lab, the temperature T and humidity H must satisfy $T \geq 20$ and $H \leq 60$. What is the feasible region?

- (A) upper left corner of the coordinate plane
 (B) lower right corner of the coordinate plane
 (C) upper right corner of the coordinate plane
 (D) lower left corner of the coordinate plane
 (E) all four corners of the coordinate plane

Q3. What is the solution to the system $2x + 3y \geq 12$ and $x - 2y \leq -3$?

- (A) all points on the line $2x + 3y = 12$
 (B) all points on the line $x - 2y = -3$
 (C) all points above the line $2x + 3y = 12$
 (D) all points below the line $x - 2y = -3$
 (E) all points above the line $2x + 3y = 12$ and below the line $x - 2y = -3$

Q4. A spacecraft requires $x \geq 5$ and $y \leq 10$ to navigate safely. Which point satisfies these conditions?

- (A) (3, 12)
 (B) (5, 10)
 (C) (3, 8)
 (D) (7, 12)
 (E) (5, 5)

Q5. The gravitational force F between two objects is given by

$F \leq \frac{Gm_1m_2}{r^2}$. If $G = 6.674 \times 10^{-11}$, $m_1 = 5.972 \times 10^{24}$, and $m_2 = 7.348 \times 10^{22}$, what is the maximum gravitational force?

- (A) $F \leq 1.99 \times 10^{16}$
 (B) $F \leq 1.99 \times 10^{20}$
 (C) $F \leq 1.99 \times 10^{22}$
 (D) $F \leq 1.99 \times 10^{24}$
 (E) $F \leq 1.99 \times 10^{26}$

Q6. What is the equation of the boundary line for the inequality $3x - 2y \geq 5$?

- (A) $3x - 2y = 5$
 (B) $3x + 2y = 5$
 (C) $2x - 3y = 5$
 (D) $2x + 3y = 5$
 (E) $x - y = 5$

Q7. A physicist is modeling the trajectory of a projectile with the inequalities $y \geq x - 2$ and $y \leq -x + 4$. What is the solution region?

- (A) the region above the line $y = x - 2$
 (B) the region below the line $y = -x + 4$
 (C) the region between the lines $y = x - 2$ and $y = -x + 4$
 (D) the region outside the lines $y = x - 2$ and $y = -x + 4$
 (E) the region below the line $y = x - 2$ and above the line $y = -x + 4$

Q8. Which inequality represents the region below the line $y = 2x - 3$?

- (A) $y \geq 2x - 3$
 (B) $y \leq 2x - 3$
 (C) $y = 2x - 3$
 (D) $y > 2x - 3$
 (E) $y < 2x - 3$

Open Ended Questions (FRQ)

Q9. Graph the system of linear inequalities $x \geq 2$ and $y \leq -x + 3$. Identify the feasible region and provide the coordinates of the vertices.

Q10. Solve the system $x + 2y \geq 4$ and $3x - 2y \leq 5$. Provide the solution region and the equations of the boundary lines.

Chef's Tips

- Tip 1: Ensure students understand the concept of feasible regions.
- Tip 2: Encourage students to visualize the solution regions.
- Tip 3: Remind students to check their work by graphing the inequalities.